



PHYSICS

What is the full QED Lagrangian with physics units written out?

Asked 3 years, 3 months ago Modified 4 months ago Viewed 3k times



I wonder what the [QED Lagrangian](#) would look like if you carefully write out all units of the terms and make sure they are consistent. I think there is something missing about Coulomb.

8



Can you write down the QED Lagrangian such that the physical units of all terms agree and state the units of all variables and terms?



lagrangian-formalism

quantum-electrodynamics

units

physical-constants



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edited Mar 18, 2023 at 18:07

asked Mar 18, 2023 at 12:44

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Gere

1,493 1 17 29

7 what is tripping you up here? This is a simple dimensional analysis problem. – [Prahar](#) Mar 18, 2023 at 12:58

8 People downvoting not because it's a bad question, but because they think you are too lazy to solve it by yourself or look for Griffiths' book on particle physics. – [DanielC](#) Mar 18, 2023 at 19:24

2 Answers

Sorted by: Highest score (default)



The Wikipedia article you linked to gives the action, not the Lagrangian or Lagrangian density, so I'll do the same.

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Note that since $x^0 = ct$ we have $d^4x = c dt d^3x$ and therefore

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$$\mathcal{L} = \int \frac{d^4x}{c} \mathcal{L}$$

density. Thus $\mathcal{L}$ is the quantity inside the density over the spatial coordinates.

gy times time. The Lagrangian L therefore must have units of energy, and the Lagrangian density $\mathcal{L}$ must have units of energy density.

The units of electromagnetic quantities such as e , ϕ , $\mathbf{A}$, $\mathbf{E}$, $\mathbf{B}$, A^μ , and $F^{\mu\nu}$ are the same in QED as in classical electromagnetism. Both the classical theory and the quantum theory can be done in various unit systems. I discuss four of them below.

The Minkowski metric $\eta_{\mu\nu}$ is dimensionless in Cartesian coordinates, and few people do QED calculations in anything other than Cartesian coordinates. In any case, since $\mathcal{L}$ must be a Lorentz scalar, if $\mathcal{L}$ has the right dimensions in Cartesian coordinates, it will have the right dimensions in any coordinates.

SI units

$$S = \int \frac{d^4x}{c} \left[-\frac{1}{4\mu_0} F^{\mu\nu} F_{\mu\nu} + \bar{\psi} (i\hbar c \gamma^\mu D_\mu - mc^2) \psi \right]$$

$$D_\mu \equiv \partial_\mu + \frac{ie}{\hbar} A_\mu$$

$$x^\mu \equiv (ct, \mathbf{x})$$

$$A^\mu \equiv \left(\frac{1}{c} \phi, \mathbf{A} \right)$$

Base units: kilograms, meters, seconds, amperes

Table of [quantity] = kg^a m^b s^c A^d

quantity	kg	m	s	A
S	1	2	−1	0
d^4x	0	4	0	0
c	0	1	−1	0
μ_0	1	1	−2	−2
$F^{\mu\nu}$ or $F_{\mu\nu}$	1	0	−2	−1
ψ or $\bar{\psi}$	0	−3/2	0	0
$\hbar$	1	2	−1	0
γ^μ or γ_μ	0	0	0	0
	1	0	0	0
	0	0	0	0
	0	0	1	1
	0	0	−2	−1
	0	0	0	0

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Gaussian units

$$S = \int \frac{d^4x}{c} \left[-\frac{1}{16\pi} F^{\mu\nu} F_{\mu\nu} + \bar{\psi} (i\hbar c \gamma^\mu D_\mu - mc^2) \psi \right]$$

$$D_\mu \equiv \partial_\mu + \frac{ie}{\hbar c} A_\mu$$

$$x^\mu \equiv (ct, \mathbf{x})$$

$$A^\mu \equiv (\phi, \mathbf{A})$$

Note: These are sometimes called "unrationalized" Gaussian units. The F^2 term has an extra factor of $1/4\pi$; this introduces factors of 4π into the field equations but removes them from solutions. For example, in these units the classical potential of a point charge is simply q/r .

In this system, μ_0 and ϵ_0 do not exist. This is because Gaussian units don't have a electromagnetic base unit; charge is defined in terms of the mechanical base units.

Base units: grams, centimeters, seconds

Table of [quantity] = $\text{g}^a \text{cm}^b \text{s}^c$

quantity	g	cm	s
S	1	2	−1
d^4x	0	4	0
c	0	1	−1
$F^{\mu\nu}$ or $F_{\mu\nu}$	1/2	−1/2	−1
ψ or $\bar{\psi}$	0	−3/2	0
$\hbar$	1	2	−1
γ^μ or γ_μ	0	0	0
m	1	0	0
∂_μ or ∂^μ	0	−1	0
e	1/2	3/2	−1
A^μ or A_μ	1/2	1/2	−1
$g_{\mu\nu}$ or $g^{\mu\nu}$	0	0	0

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$$(i\hbar c \gamma^\mu D_\mu - mc^2) \psi \Big]$$

units. The F^2 term has a prefactor of $-1/4$ in the field equations.

Particle physics units

$$S = \int d^4x \left[-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi \right]$$

$$D_\mu \equiv \partial_\mu + ieA_\mu$$

$$x^\mu \equiv (t, \mathbf{x})$$

$$A^\mu \equiv (\phi, \mathbf{A})$$

Base units: GeV

Table of [quantity] = GeV^a

quantity	GeV
$c \equiv 1$	0
$\hbar \equiv 1$	0
S	0
d^4x	−4
$F^{\mu\nu}$ or $F_{\mu\nu}$	2
ψ or $\bar{\psi}$	3/2
γ^μ or γ_μ	0
m	1
∂_μ or ∂^μ	1
e	0
A^μ or A_μ	1
$g_{\mu\nu}$ or $g^{\mu\nu}$	0

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edited Mar 19, 2023 at 2:43

answered Mar 18, 2023 at 21:16

 **Ghoster**
995 ● 2 ● 14 ● 15

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main stumbling block was μ_0 which did not
hand. Do you know which book would mention

though. For that you would need to fix units such
Do you know a reference discussing how to make

article on the [covariant formulation of classical](#)
7:16

- 2 This makes Coulomb's Law be $F = q_1 q_2 / r^2$, so there is no silly $1/4\pi\epsilon_0$ as in SI. Similarly, there is no need for a μ_0 in the force between two currents. – [Ghoster](#) Mar 19, 2023 at 17:26
- 1 Actually, the $-\frac{1}{4\mu_0}$ is $-\frac{1}{4}\epsilon_0 c \sqrt{|g|}$. There is a $\sqrt{|g|}$ factor there, and for all the other terms in the action integral. In addition, the dimensions for the components $F_{\mu\nu}$, $g_{\mu\nu}$, ∂_μ , d^4x , etc. depend on what type of coordinates you're using (e.g. $[\partial_\mu] = 1$ if x^μ is an angular coordinate). In particular, d^4x can have all sorts of different dimensions (e.g. TL for spherical coordinates and $x^0 = t$). Instead, it's $\sqrt{|g|} d^4x$ that has the consistent dimension (L^4), where L denotes length, T time. – [NinjaDarth](#) Oct 11, 2023 at 16:39



With a gauge-fixing term added in:

2

$$S = \int \left(-\frac{\epsilon_0 c}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma} - \frac{\epsilon_0 c}{2} (g^{\mu\nu} \partial_\mu A_\nu)^2 + \bar{\psi} \left(\gamma^\mu (i\hbar \overleftrightarrow{\partial}_\mu + eA_\mu) - mc \right) \psi \right) \sqrt{|g|} d^4x,$$



where



$$\epsilon_0 \mu_0 = \frac{1}{c^2}, \quad \overleftrightarrow{\partial}_\mu \equiv \frac{\overrightarrow{\partial}_\mu - \overleftarrow{\partial}_\mu}{2}$$

with the arrows denoting the direction that the operator is to be applied in. This is to be understood as passing through the gammas, so it should actually be written as:

$$\gamma^\mu i\hbar \overleftrightarrow{\partial}_\mu = i\hbar \frac{\gamma^\mu \overrightarrow{\partial}_\mu - \overleftarrow{\partial}_\mu \gamma^\mu}{2}.$$

Dimensions ... set:

$$[\mu] = [x^\mu], \quad \Omega = [d^4x].$$

Then

$$[dx^\mu] = [\mu], \quad \Omega = [0][1][2][3], \quad [\partial_\mu] = \frac{1}{[\mu]} = \left[\overleftrightarrow{\partial}_\mu \right].$$

For the linear-operators:

$$[g(-, -)] = L^2, \quad [g^{-1}(-, -)] = \frac{1}{L^2},$$

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$$[g^{\mu\nu} \equiv g^{-1}(dx^\mu, dx^\nu)] = \frac{[\mu][\nu]}{L^2}.$$

ly dimensionless! Their dimensions are

$$\left[g_{\theta\theta} \equiv g \left(\frac{\partial}{\partial\theta}, \frac{\partial}{\partial\theta} \right) = r^2 \right] = L^2,$$

for spherical coordinates;

$$\left[g_{tt} \equiv g \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial t} \right) \right] = \frac{L^2}{T^2},$$

since $[\theta] = 1$ and $[t] = T$.

And yes: **there is a $\sqrt{|g|}$ factor in the Lagrangian density**, because it's a scalar density, not a scalar! And its dimension is coordinate-dependent. Example: for spherical coordinates and $x^0 = t$, $\Omega = [0][1][2][3] = T \cdot L \cdot 1 \cdot 1 = TL$, and $[\sqrt{|g|}] = L^4/(TL) = L^3/T$.

For mass and (in vacuo) light speed, Planck's constant and action, respectively:

$$[m] = M, \quad [c] = \frac{L}{T}, \quad [\hbar] = H \equiv \frac{ML^2}{T} = [S],$$

where M denotes dimension of mass, T of time, and H of action.

For the charge and electromagnetic field

$$[e] = Q, \quad [A_\mu] = \frac{P}{[\mu]}, \quad [F_{\mu\nu}] = \frac{P}{[\mu][\nu]}, \quad [\varepsilon_0] = \frac{QT}{PL}, \quad [\mu_0] = \frac{PT}{QL}, \quad PQ = H,$$

where Q denotes the dimension of electric charge and P the conjugate dimension of magnetic charge/flux.

Finally, for the spinor, the gamma linear operator and its components:

$$[\gamma(-)] = \frac{1}{L}, \quad [\gamma^\mu \equiv \gamma(dx^\mu)] = \frac{[\mu]}{L}, \quad [\bar{\psi}] = \sqrt{\frac{1}{L^3}} = [\psi].$$

And, again: **the gamma matrices are not generally dimensionless**, but are coordinate-dependent; example:

$$\left[\gamma^t \equiv \gamma(dt) = \frac{\gamma^0}{c} \right] = \frac{[dt]}{L} = \frac{T}{L}, \quad \left[\gamma^\theta \equiv \gamma(d\theta) = \frac{\gamma^2}{r} \right] = \frac{[d\theta]}{L} = \frac{1}{L},$$

since the components that multiply dx^μ are associated with orthonormal co-frames,

$$(rd\theta), \quad \gamma^3 = \gamma(r \sin \theta d\varphi),$$

$$\frac{\gamma^2}{r}, \quad \gamma^\varphi = \frac{\gamma^3}{r \sin \theta}.$$

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Finally... the situation with the fractional dimensions for $\bar{\psi}$ and ψ is a clear tell - a hallmark sign - that there's conflation going on, just like with the Gaussian units for electromagnetism, and that things are getting mixed in together and confused that should be kept separate. So, actually, we should be cutting the moorings loose here, too, like we do with the invocation of Q and P for electromagnetism, and set:

$$[\bar{\psi}] = F = [\psi],$$

where F denotes the newly-invoked dimension of "**fermionity**". That means modifying the Dirac part of the action to:

$$S = \int \left(\cdots + \bar{\psi} \left(F_0 \gamma^\mu (i\hbar \overleftrightarrow{\partial}_\mu + eA_\mu) - F'_0 mc \right) \psi \right) \sqrt{|g|} d^4x,$$

with two "**vacuum fermittivity**" coefficients with respective dimensions:

$$[F_0] = \frac{1}{L^3 F^2} = [F'_0].$$

Then, the renormalized Lagrangian densities, are of the very same form, except for the replacements:

$$\varepsilon_0 \rightarrow Z_3 \varepsilon_R, \quad \left(\mu_0 \rightarrow \frac{\mu_R}{Z_3} \right), \quad F_0 \rightarrow Z_1 F_R = Z_2 F_R, \quad F'_0 = Z_0 F'_R.$$

It is not the fields and charges that are "bare" versus "dressed", but the coefficients themselves. It's the vacuum that's being regularized and renormalized, and it is what suffers the "bare" versus "dressed" distinction.

The Z 's, themselves, are the shadows of explicit or tacit coefficients, like $(\varepsilon_0, \mu_0, F_0, F'_0)$, that is also a tell which shows that there should be coefficients there.

For reference, by the way:

$$\mu_0 c = \ll Z_0 \gg = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{1}{\varepsilon_0 c},$$

is the "impedance of free space" and has a more less standard symbol all its own - the $\ll Z_0 \gg$ I put in scare quotes because it's a different Z_0 than the renormalization coefficient: a name conflict.

But I think that might have been the case if Z for the renormalization coefficients themselves

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and the electromagnetic part of the action is:

$$S = \int \left(-\frac{g^{\mu\rho}g^{\nu\sigma}F_{\mu\nu}F_{\rho\sigma}}{4\ll Z_0 \gg} - \frac{(g^{\mu\nu}\partial_\mu A_\nu)^2}{2\ll Z_0 \gg} + \dots \right) \sqrt{|g|}d^4x.$$

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edited Oct 11, 2023 at 16:46

answered Oct 10, 2023 at 18:14

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NinjaDarth

3,091 ● 8 ● 13

- 1 That's really good - especially the careful attention to details, dotting the i's and crossing the t's. But a table might help, because it's all spread out! – [Lydia Marie Williamson](#) Oct 11, 2023 at 0:14
- 1 I might put in something for "power counting" - making particular note that it's not necessarily the coordinates that contribute to it (since they can be angular), but the $\sqrt{|g|}$ that does. So, it's not d^4x that generally has a power count of -4, but $\sqrt{|g|} d^4x$ that does! – [NinjaDarth](#) Oct 11, 2023 at 16:53

Thanks, also for the "dot-i's and cross-t's" expression. I think I'll be using that more often. – [NinjaDarth](#) Dec 16, 2023 at 1:48

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